

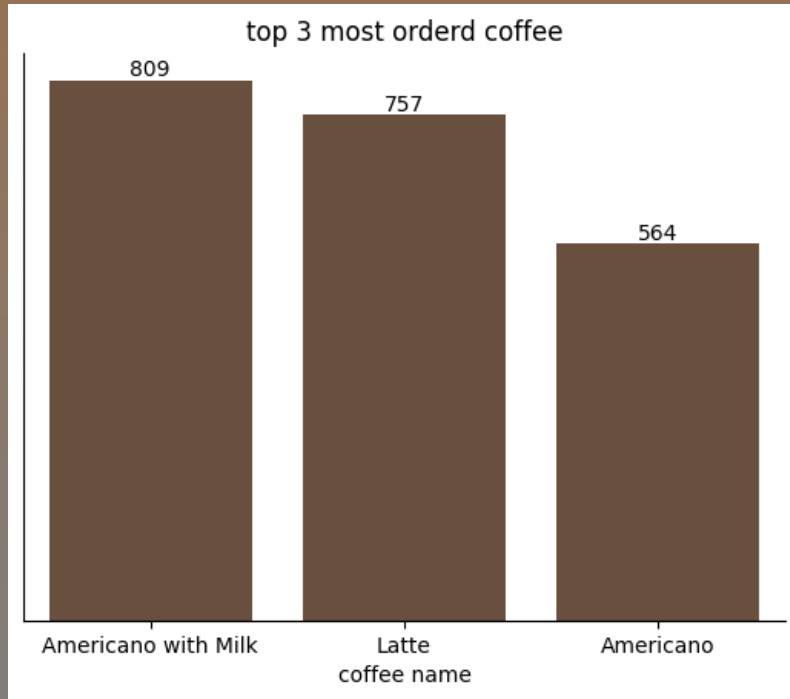
A high-quality photograph of a white ceramic cup filled with dark coffee, topped with a thin layer of foam. The cup sits on a matching white saucer. Both are surrounded by a large pile of dark brown, roasted coffee beans. In the upper right corner, a wooden scoop is partially visible, also filled with beans. The entire scene is set against a light brown, textured background, possibly burlap or a similar natural fabric.

Time Café Analysis

Tools : Python , Power Point

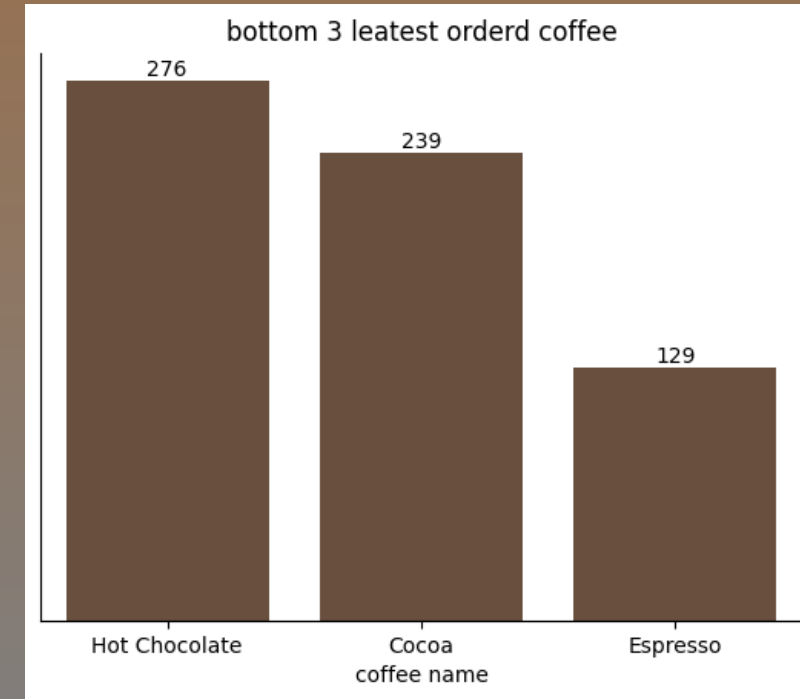
Maid by : Abrar Mahmoud

The top 3 products sold



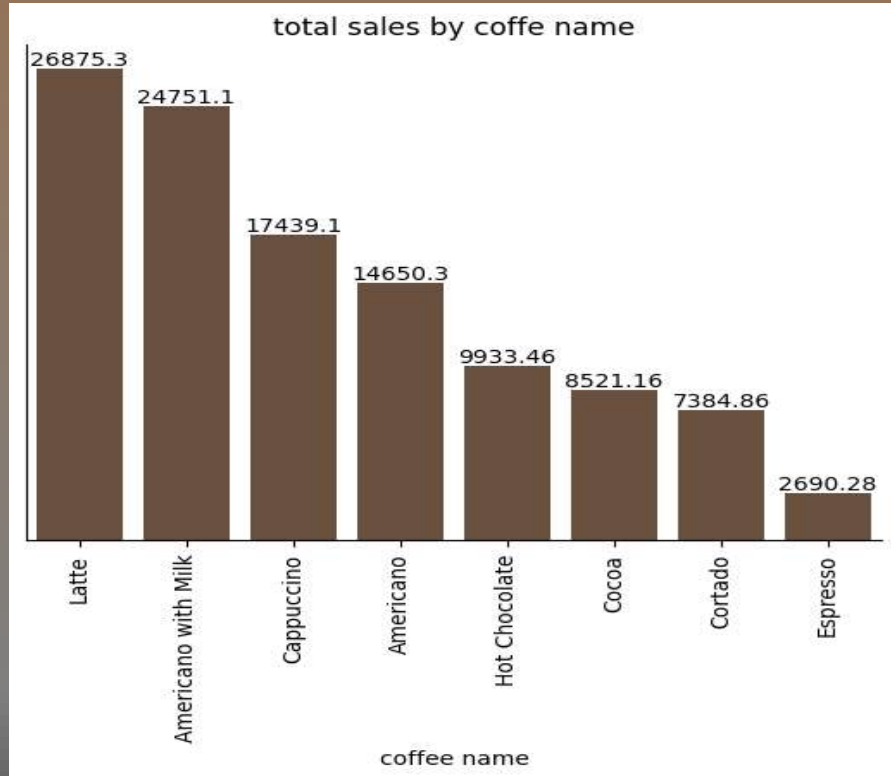
- Americano with Milk had the highest number of orders (809).
- Latte ranked second (757), followed by Americano (564).
- Milk-based coffee drinks are the most popular among customers.

The 3 least sold products



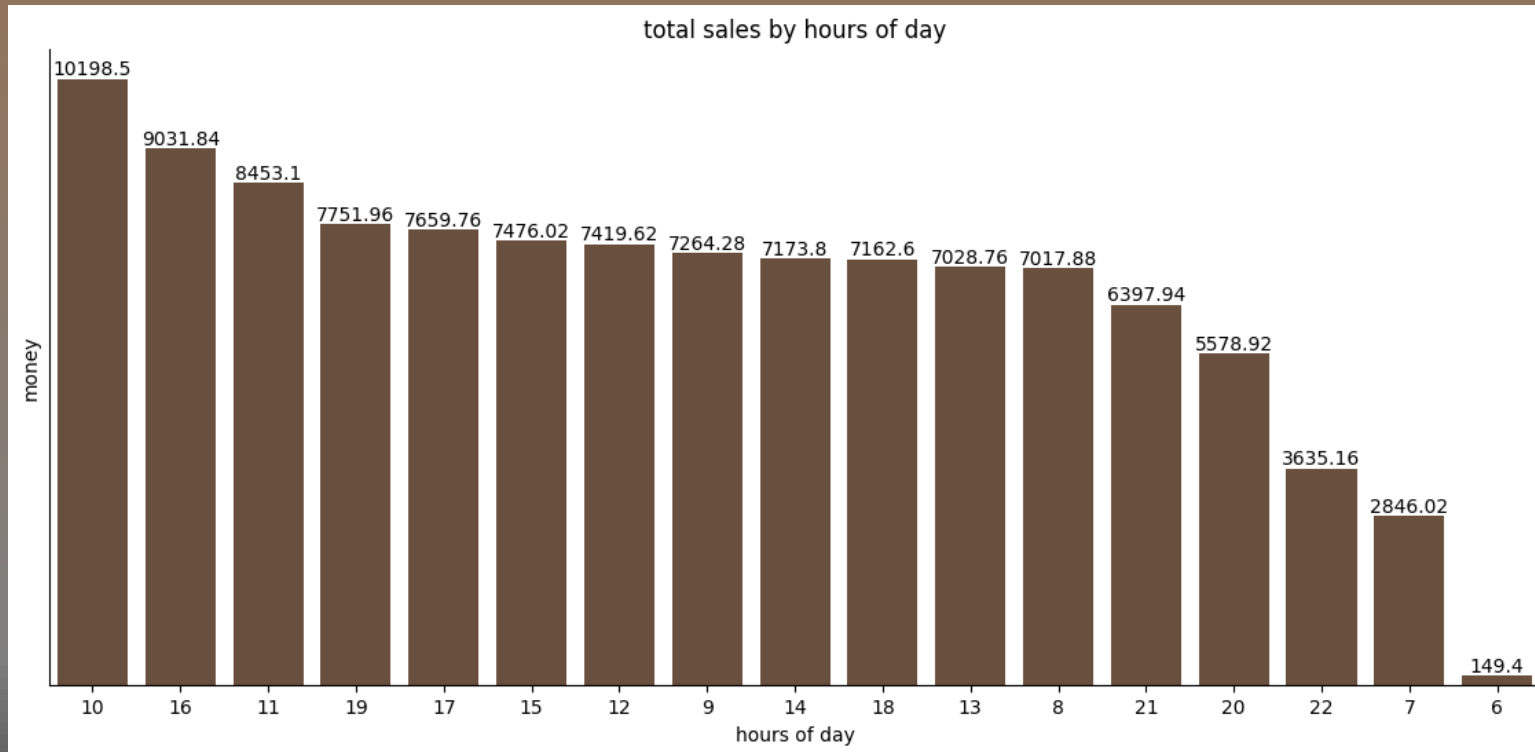
- Espresso had the lowest number of orders (129).
- Cocoa (239) and Hot Chocolate (276) also showed low demand.
- These drinks may need promotions or menu improvements to increase sales.

Which drink makes the most revenue?



- Latte generated the highest total sales (26,875.3), followed by Americano with Milk (24,751.1).
- Espresso had the lowest total sales (2,690.3).
- Popular drinks contribute the most to overall revenue.

Is there a difference between "the most demanded" and the "most profitable"?



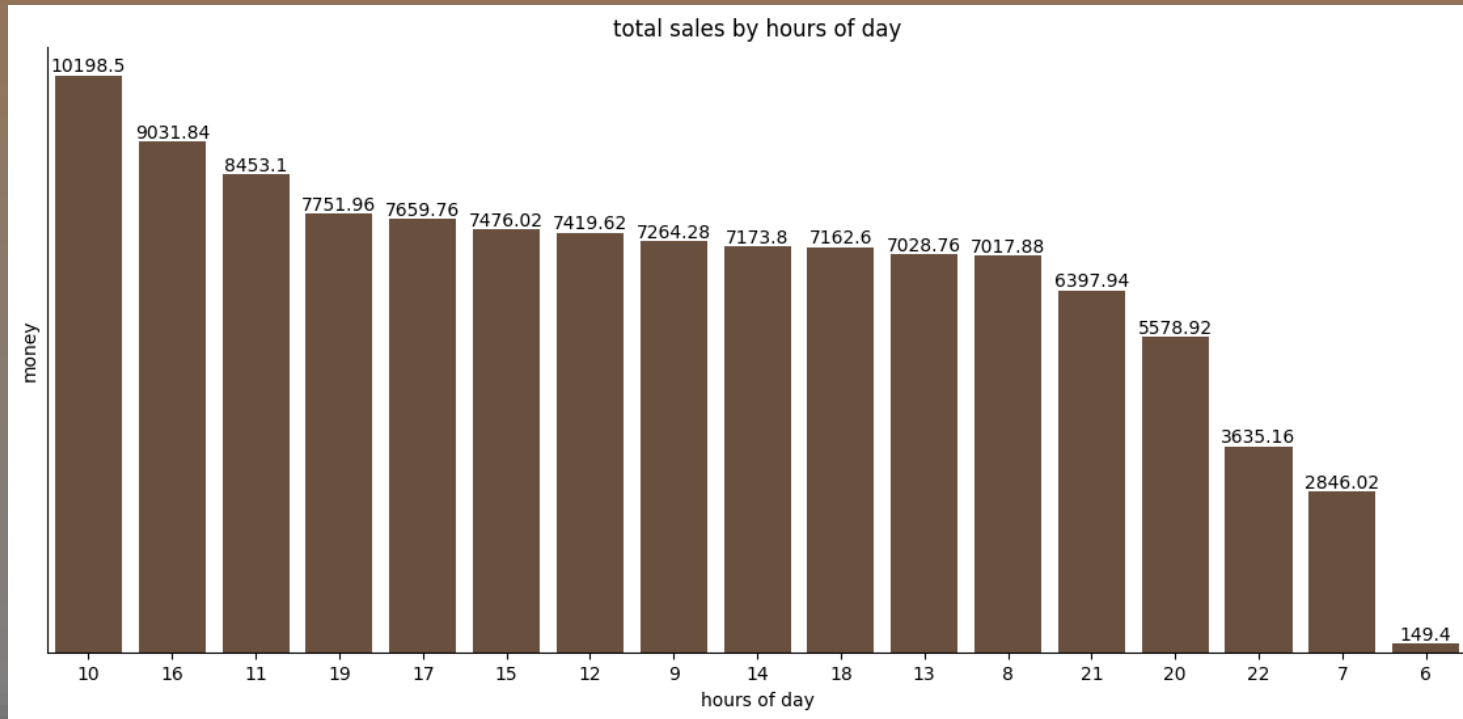
- **The Short Answer:** Yes, they are different.
- What the Chart Shows (Profitability): 10 AM is the winner (\$10.2K).
- What the Chart Hides (Demand): Total number of orders.
- Strategic Takeaway: High revenue $\neq$ High traffic. 10 AM's sales might just be a few expensive orders, not actual high demand.
- Next Step to Decide Focus: Pull an "Order Volume by Hour" report.

Distribution of the price of each drink

	mean	sum	max	min
coffee_name				
Americano	25.975638	14650.26	28.9	23.02
Americano with Milk	30.594710	24751.12	33.8	27.92
Cappuccino	35.883004	17439.14	38.7	32.82
Cocoa	35.653389	8521.16	38.7	32.82
Cortado	25.731220	7384.86	28.9	23.02
Espresso	20.854884	2690.28	24.0	18.12
Hot Chocolate	35.990797	9933.46	38.7	32.82
Latte	35.502378	26875.30	38.7	32.82

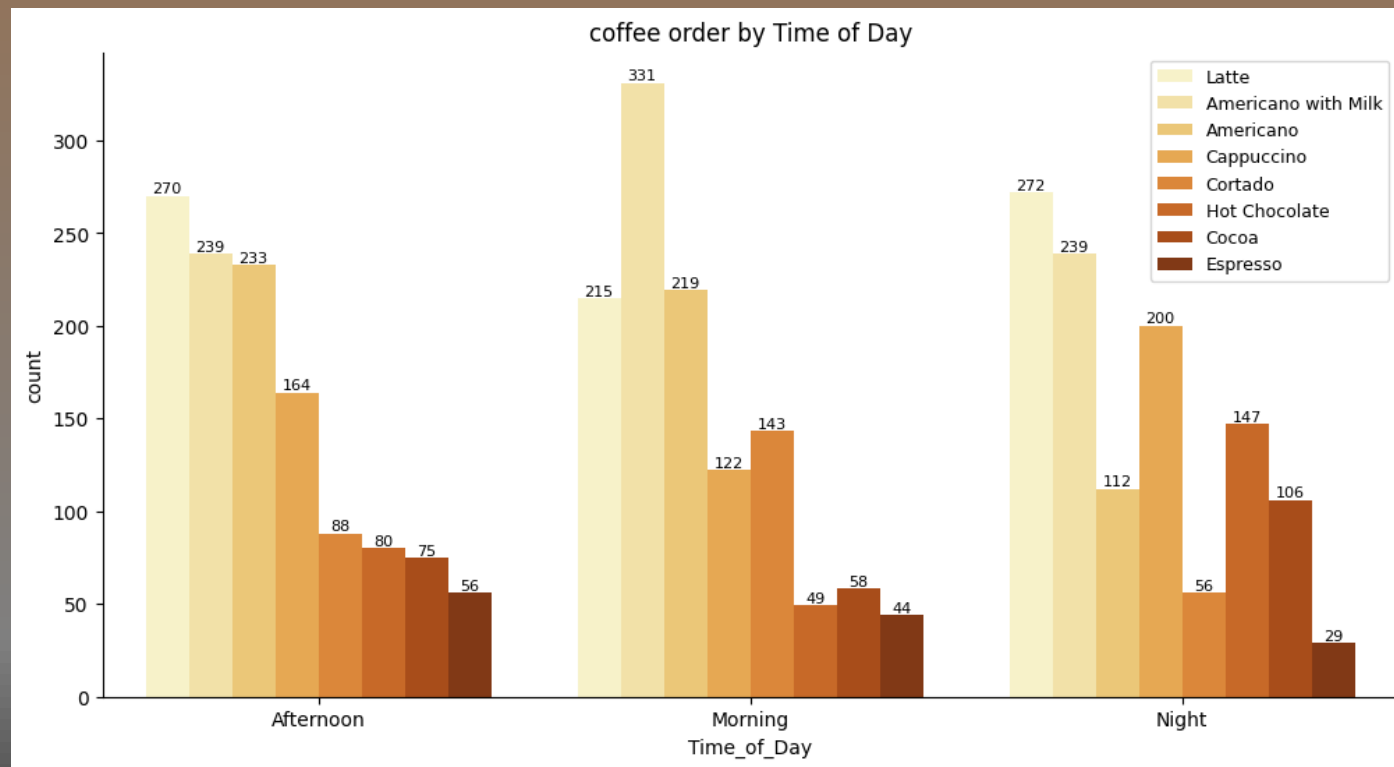
- Metrics Tracked: Actual Min, Max, and Average price per drink.
- Key Observation: Noticeable price gaps between the "Min" and "Max" for almost every item (e.g., Americano averages ~\$30, but fluctuates).
- What this Means: Customers aren't just buying standard drinks; they are actively customizing (sizes, extra shots, syrups) or applying discounts.
- Strategic Takeaway: Drinks with the biggest gap between Average and Max price are your best upselling opportunities.

Which hour of the day has the most sales?



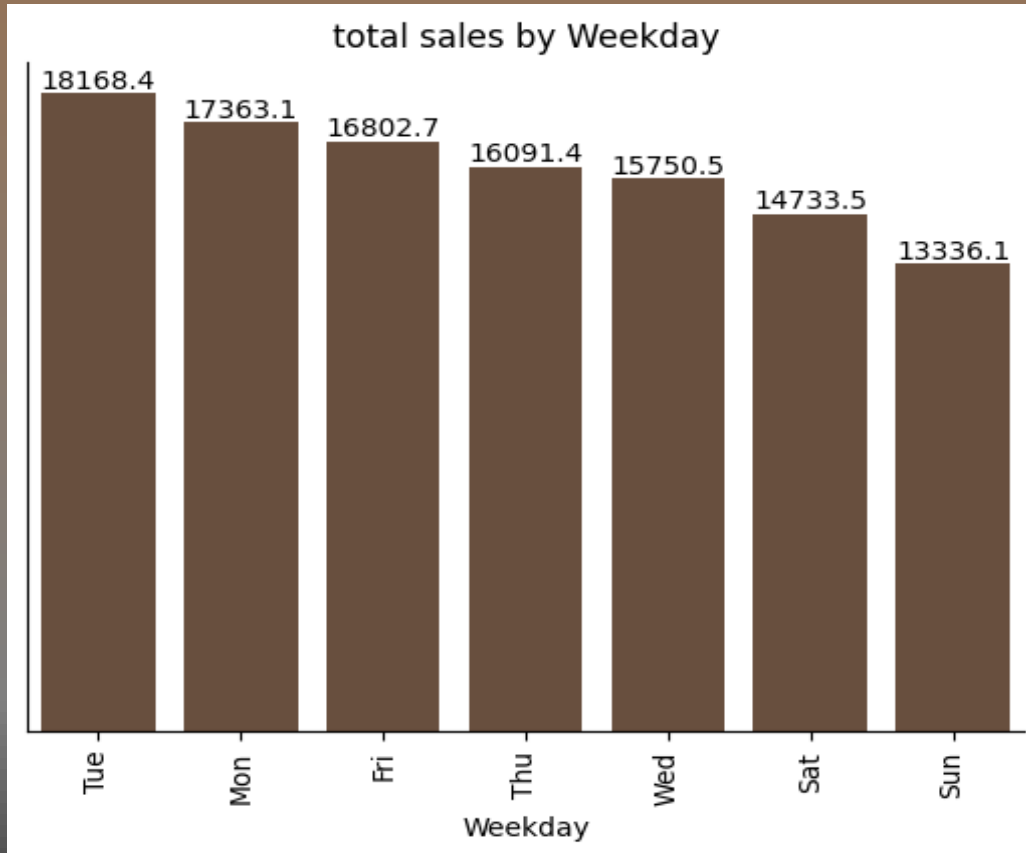
- Most Crowded: 10:00 AM (\$10.2K Revenue)
- Slowest Hour: 6:00 AM (\$149 Revenue)
- Staffing Strategy: Schedule maximum staff at 10 AM. Apply a "scaling down" shift model after 11 AM to match the steady drop in sales.

Are sales in the morning different from the afternoon and evening in terms of quantity and the type of drink ordered?



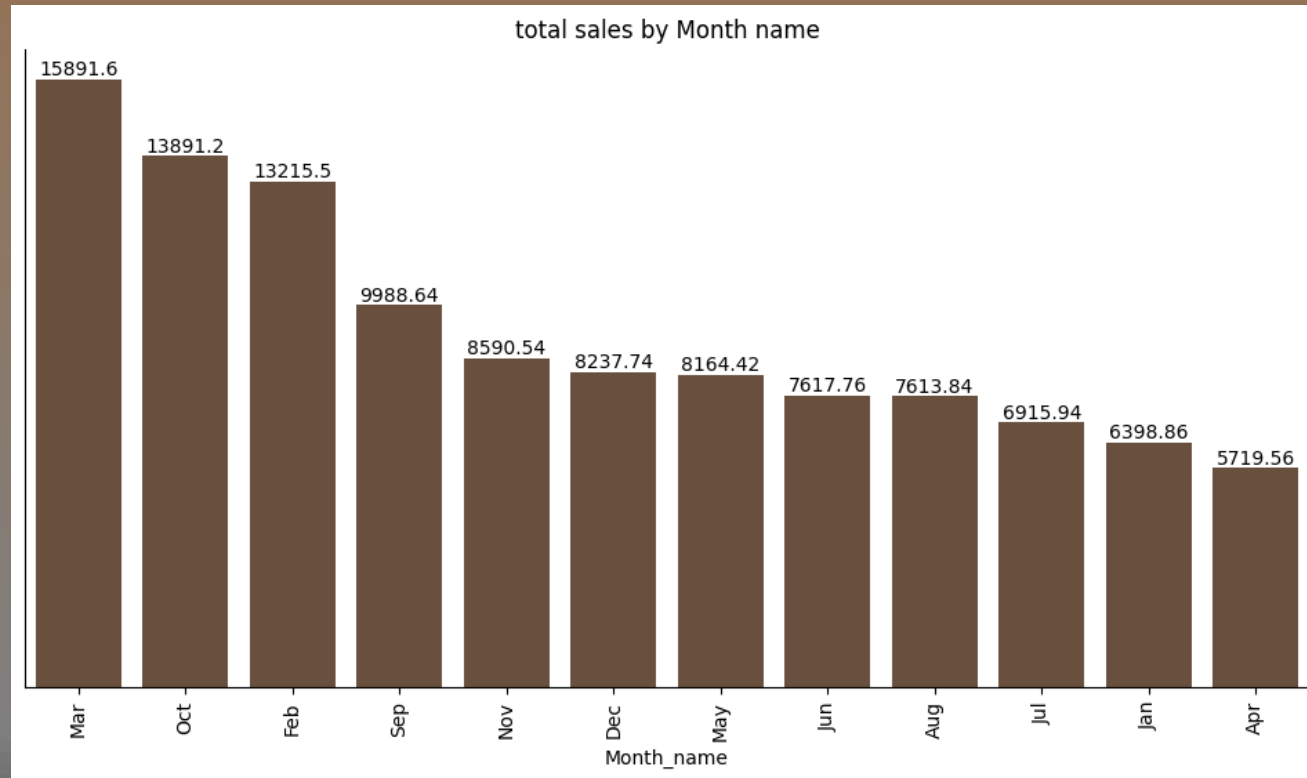
- Quantity (Volume): Morning has the lowest order count for top sellers (e.g., Latte: 215) compared to Afternoon (270) & Night (272).
- Drink Type: Latte & Americano dominate all day.
- Behavior Shift: Morning is the only time Americano almost matches Latte. Afternoon & Night show a strong shift towards Lattes.
- Strategic Insight: Morning brings fewer orders but higher revenue (ties back to 10 AM peak = likely expensive/customized orders, not high traffic).

Which days of the week sell the most? And is the weekend different from the weekdays?



- Peak Day: Tuesday (\$18.1K)
- Lowest Day: Sunday (\$13.3K)
- Weekday vs. Weekend: Yes, a major difference. Weekends (Sat & Sun) are the lowest performing days.
- Clear Trend: Sales peak early in the week and steadily decline towards Sunday.

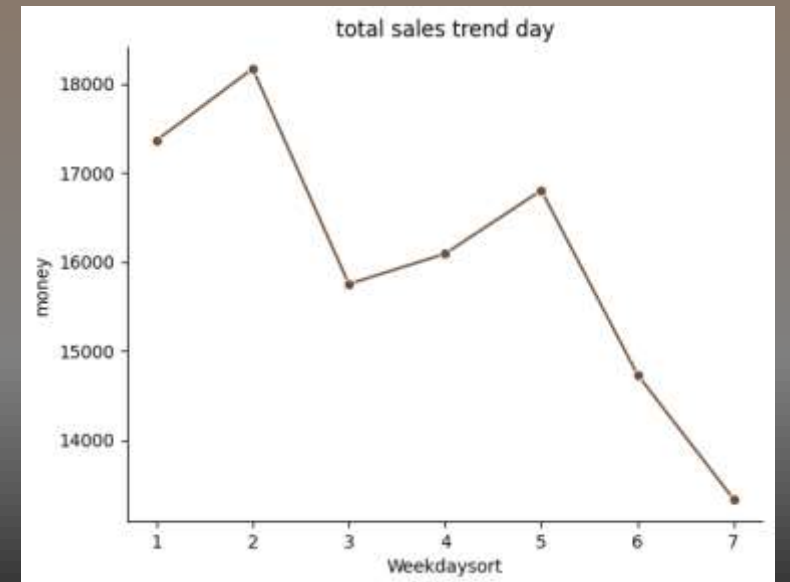
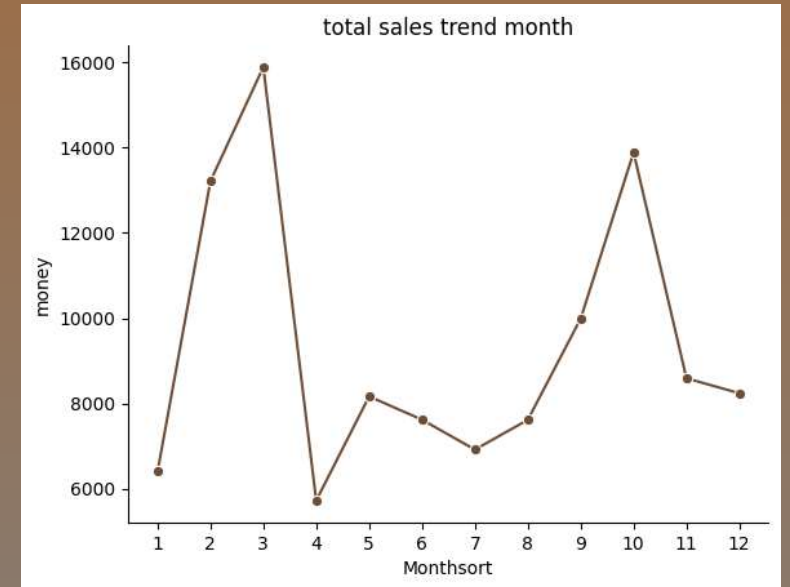
Are there months with higher sales than others? Could it be related to the weather or events?



- Peak Months: March (\$15.9K) & October (\$13.9K)
- Lowest Month: April
- The "Weather" Factor: Top sales perfectly align with transitional weather (Spring & Autumn). Summer months (Jul/Aug) show a clear drop, likely due to lower hot coffee demand.
- The "Event" Anomaly: December (Holidays) performs mid-to-low—no holiday sales spike observed here.

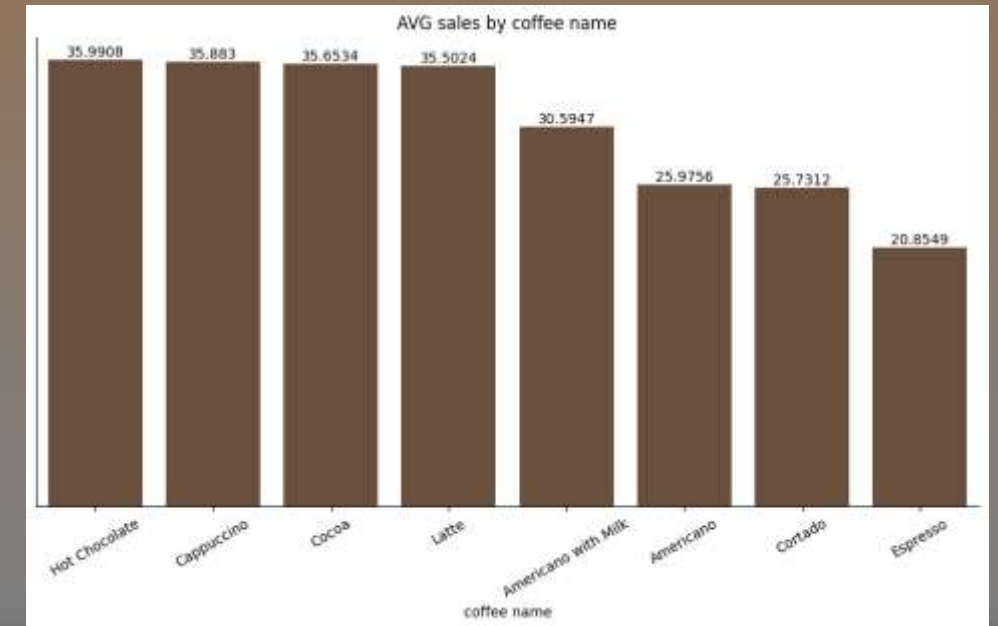
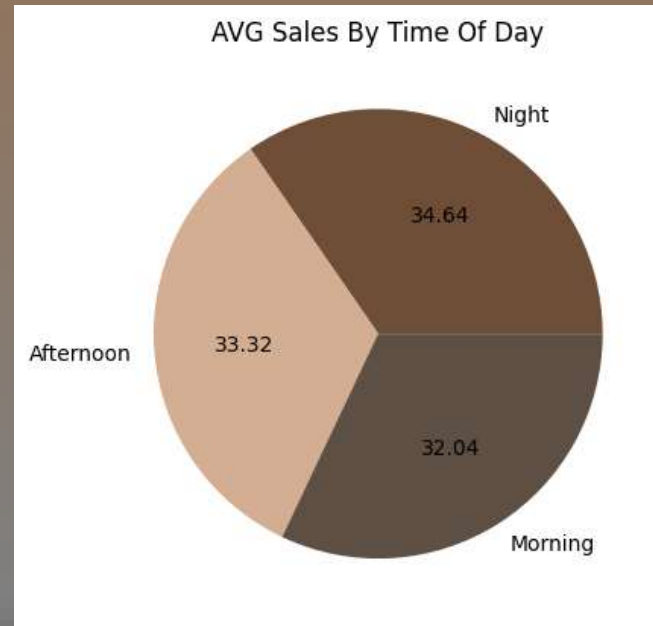
The total daily and monthly revenue, and is there a trend of increase or decrease?

- Monthly Trend: Highly volatile. No steady growth or decline. (Spikes in March & October, sudden crash in April).
- Daily Trend: Clear downward slope. Peaks early in the week (Days 1 & 2 at ~\$18K) and drops steadily to Day 7 (~\$13K).
- Conclusion: Monthly revenue is unpredictable (likely season/event-driven), but the weekly downward trend is highly consistent (weekend slump).



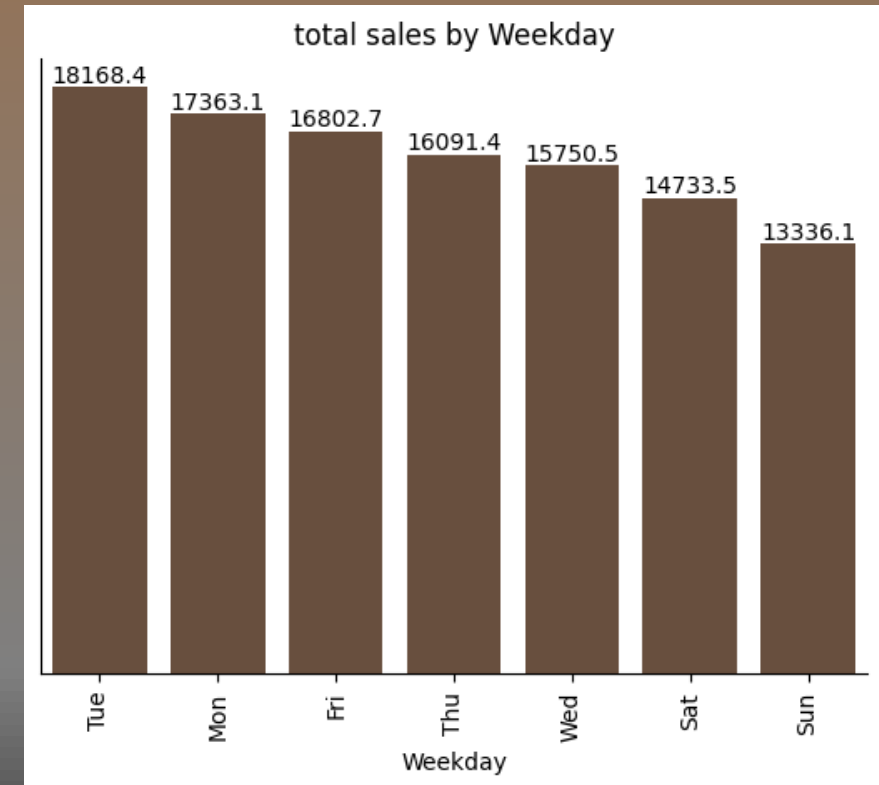
The average value of a single bill, and does it change depending on the time or the drink?

- Average Transaction Value (Basket Size)
- By Time of Day: Virtually no difference.
- Night: 34.6%
- Afternoon: 33.3%
- Morning: 32.0%
- **Key Takeaway:** Customers spend the exact same average amount per order, regardless of when they visit.



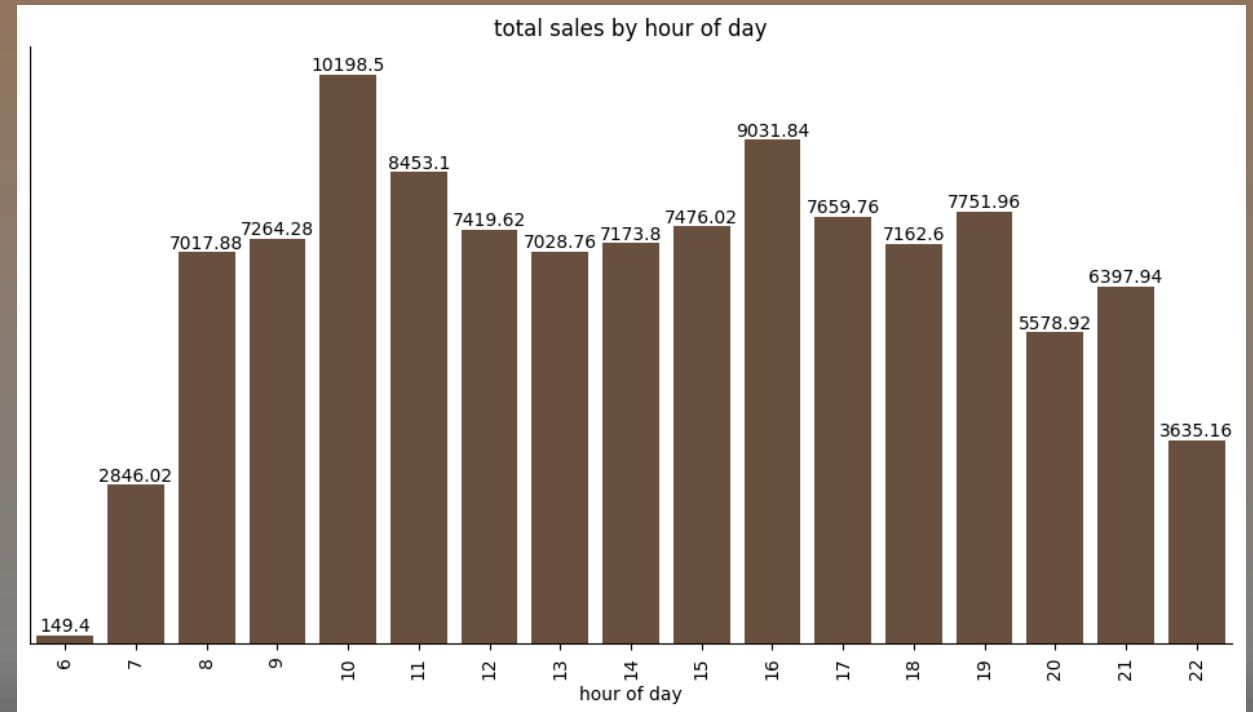
Are there certain days or hours with noticeably higher revenue that are worth focusing on with promotions?

- The Chart Reality: This chart shows what sells for the most, not when.
- Highest Ticket Items: Hot Chocolate (\$36), Cappuccino (\$35.8), Cocoa (\$35.6).
- Lowest Ticket Items: Espresso (\$20.8), Cortado (\$25.7).
- Strategic Shift: Instead of just offering time-based discounts, focus your offers on high-value items.
- Actionable Offer: Bundle Hot Chocolate, Cappuccino, or Cocoa with food during slow hours (Afternoons/Weekends) to maximize order value.



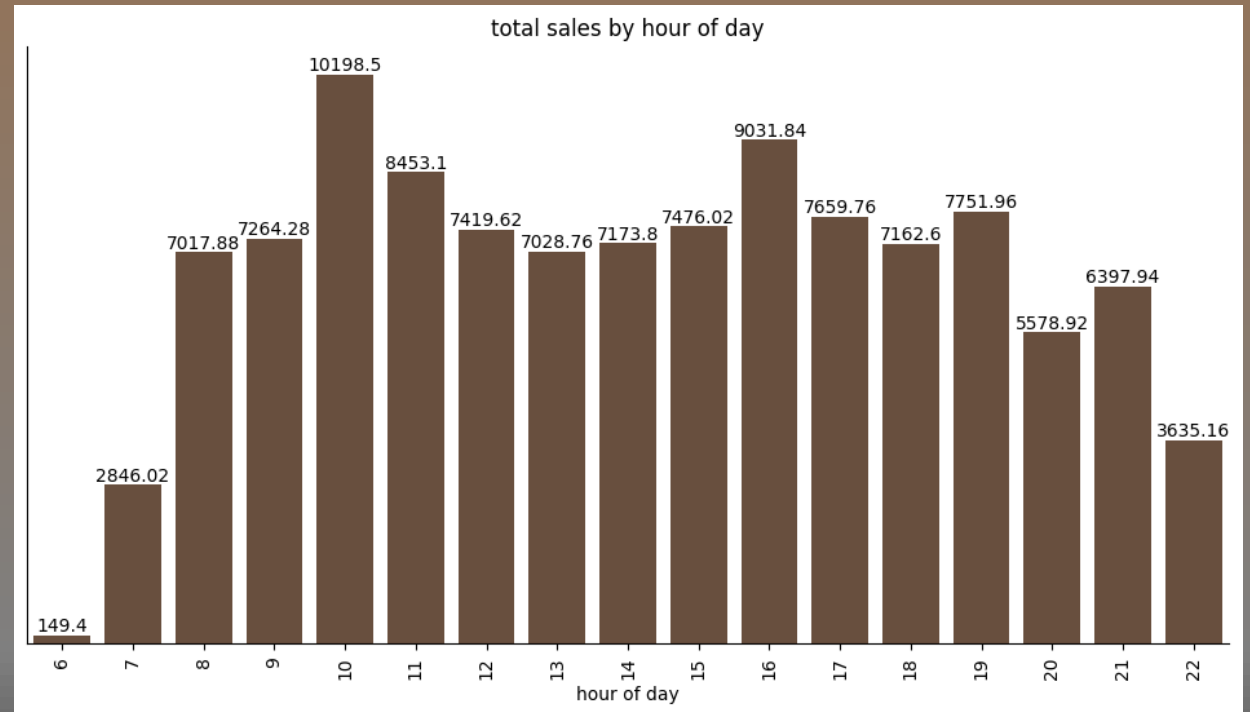
A simple count of the number and percentage of each payment method.

- Sales Analysis by Hour of the DaySales activity starts very low at 6:00 AM, then increases significantly from 8:00 AM onward. The highest sales were recorded at 10:00 AM, reaching approximately 10,198.5, indicating the strongest customer activity during the morning.A second notable sales peak occurred at 4:00 PM with around 9,031.8, suggesting another busy shopping period.
- Sales remained relatively stable between 11:00 AM and 7:00 PM, averaging between 7,000–8,500.After 8:00 PM, sales declined noticeably, reaching their lowest evening level by 10:00 PM.
- **Key Insight :** The business experiences two major peak periods—10:00 AM and 4:00 PM. These time windows are ideal for scheduling promotions, ensuring adequate staffing, and maintaining inventory availability.



A simple count of the number and percentage of each payment method.

- Payment Method Analysis 100% of all transactions were completed using Card payments. No transactions were recorded using other payment methods such as Cash or Digital Wallets.
- **Key Insight :** The dataset indicates complete reliance on card payments, which may reflect the business model, payment policy, or the scope of the available data. This is an important finding to highlight, as it may influence customer behavior analysis and payment strategy decisions.





THANK YOU